

Initiating Search

November 6, 2025, 5:11 PM

• Search:

CAS Lexicon Search:

Diisobutylaluminum hydride - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (5,160)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (368)	 Reactions	View Results
Filtered By:		
Document Type:	Journal	

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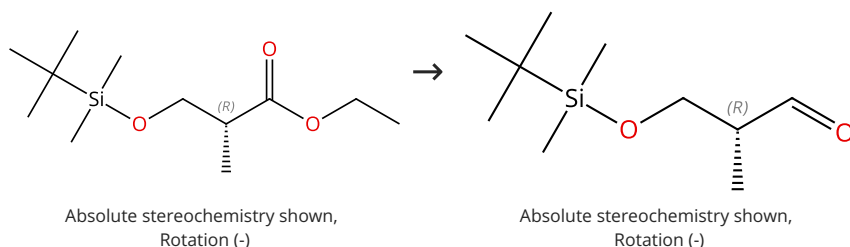
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (5)

31-517-CAS-8395317

Steps: 1 Yield: 100%

Diisobutylaluminum Hydride Reductions Revitalized: A Fast, Robust, and Selective Continuous Flow System for Aldehyde Synthesis

By: Webb, Damien; et al

Organic Letters (2012), 14(2), 568-571.

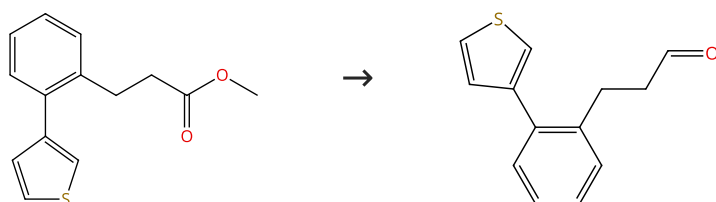
1.1 Reagents: Diisobutylaluminum hydride
Solvents: Toluene; 13.9 s, -78 °C

1.2 Reagents: Methanol

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-517-CAS-7879419

Steps: 1 Yield: 100%

Mechanistic Insights into the Post-Cyclization Isomerization in Gold-Catalyzed 7-exo-dig-Hydroarylations

By: Pflaesterer, Daniel; et al

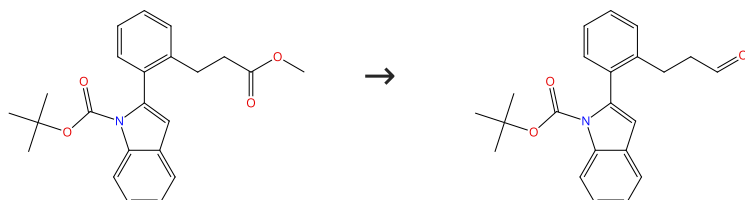
Chemistry - A European Journal (2015), 21(32), 11585-11589.

1.1 Reagents: Diisobutylaluminum hydride
Solvents: Dichloromethane, Hexane; 2 h, -90 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



31-517-CAS-2985408

Steps: 1 Yield: 100%

Mechanistic Insights into the Post-Cyclization Isomerization in Gold-Catalyzed 7-exo-dig-Hydroarylations

By: Pflaesterer, Daniel; et al

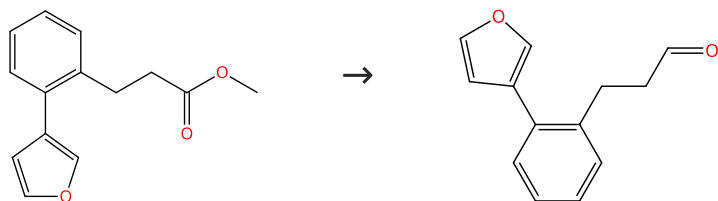
Chemistry - A European Journal (2015), 21(32), 11585-11589.

1.1 Reagents: Diisobutylaluminum hydride
Solvents: Dichloromethane, Toluene; 2 h, -90 °C

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



31-517-CAS-13763508

Steps: 1 Yield: 100%

Mechanistic Insights into the Post-Cyclization Isomerization in Gold-Catalyzed 7-exo-dig-Hydroarylations

By: Pflaesterer, Daniel; et al

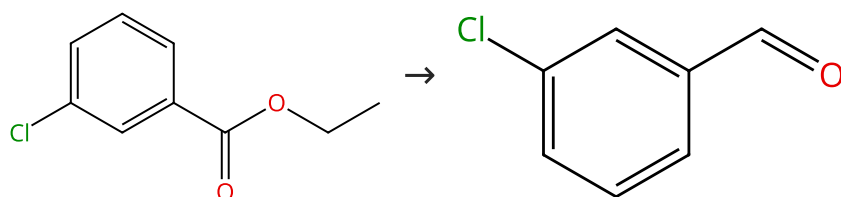
Chemistry - A European Journal (2015), 21(32), 11585-11589.

1.1 **Reagents:** Diisobutylaluminum hydride
Solvents: Dichloromethane, Hexane; 2 h, -90 °C

Experimental Protocols

Scheme 5 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (67)

Suppliers (89)

31-614-CAS-33829818

Steps: 1 Yield: 99%

Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives

By: Young Kim, So; et al

ChemistrySelect (2022), 7(30), e202202351.

1.1 **Reagents:** Morpholine, Diisobutylaluminum hydride
Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C

1.2 10 min, 0 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water; 0 °C

Experimental Protocols

31-517-CAS-11181320

Steps: 1 Yield: 99%

Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters

By: Jeon, Ah Ram; et al

Tetrahedron (2014), 70(29), 4420-4424.

1.1 **Reagents:** Morpholine, Diisobutylaluminum hydride
Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C

1.2 0 °C; 10 min, 0 °C

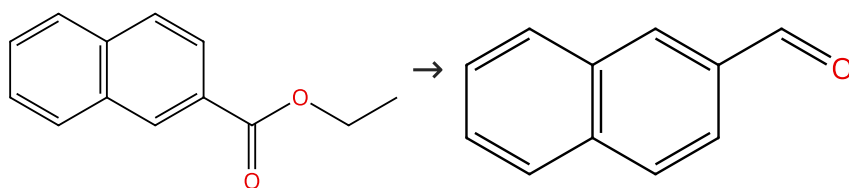
1.3 **Reagents:** Diisobutylaluminum hydride
Solvents: Hexane; 10 min, 0 °C

1.4 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 6 (2 Reactions)

Steps: 1 Yield: 99%



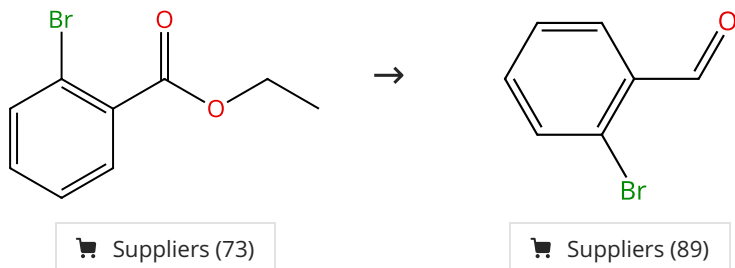
Suppliers (56)

Suppliers (86)

31-614-CAS-33829852 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C 1.2 10 min, 0 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C Experimental Protocols	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives By: Young Kim, So; et al ChemistrySelect (2022), 7(30), e202202351.
31-517-CAS-4450372 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.

Scheme 7 (2 Reactions)

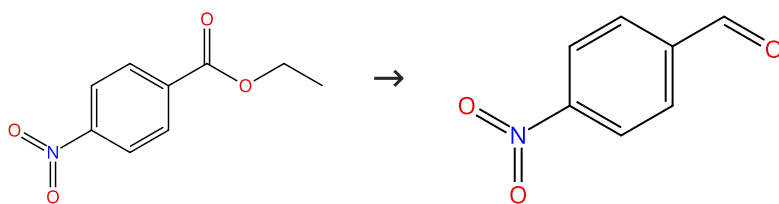
Steps: 1 Yield: 99%



31-614-CAS-33829822 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C 1.2 10 min, 0 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C Experimental Protocols	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives By: Young Kim, So; et al ChemistrySelect (2022), 7(30), e202202351.
31-517-CAS-15441094 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.

Scheme 8 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (92)

31-614-CAS-33829831

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Morpholine, Diisobutylaluminum hydride
Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C
- 1.2 10 min, 0 °C
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water; 0 °C

Experimental Protocols

Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives

By: Young Kim, So; et al

ChemistrySelect (2022), 7(30), e202202351.

31-517-CAS-4139251

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Morpholine, Diisobutylaluminum hydride
Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C
- 1.2 0 °C; 10 min, 0 °C
- 1.3 **Reagents:** Diisobutylaluminum hydride
Solvents: Hexane; 10 min, 0 °C
- 1.4 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

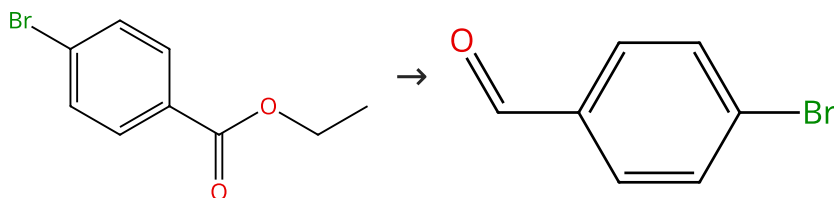
Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters

By: Jeon, Ah Ram; et al

Tetrahedron (2014), 70(29), 4420-4424.

Scheme 9 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (81)

Suppliers (91)

31-614-CAS-33829832

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Morpholine, Diisobutylaluminum hydride
Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C
- 1.2 10 min, 0 °C
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water; 0 °C

Experimental Protocols

Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives

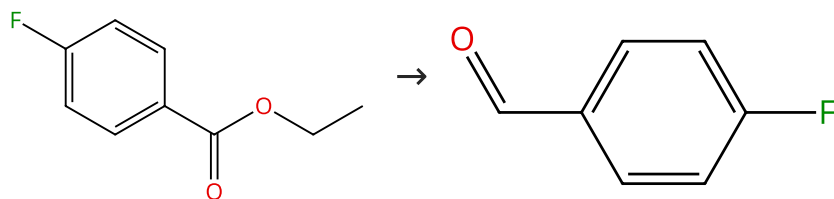
By: Young Kim, So; et al

ChemistrySelect (2022), 7(30), e202202351.

31-517-CAS-2037418 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.
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Scheme 10 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (81)

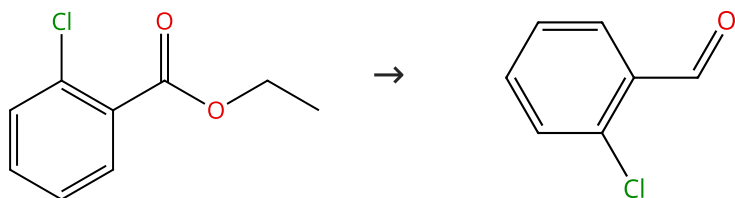
Suppliers (96)

31-614-CAS-33829811 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C 1.2 10 min, 0 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C Experimental Protocols	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives By: Young Kim, So; et al ChemistrySelect (2022), 7(30), e202202351.
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31-517-CAS-9050372 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.
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Scheme 11 (1 Reaction)

Steps: 1 Yield: 99%



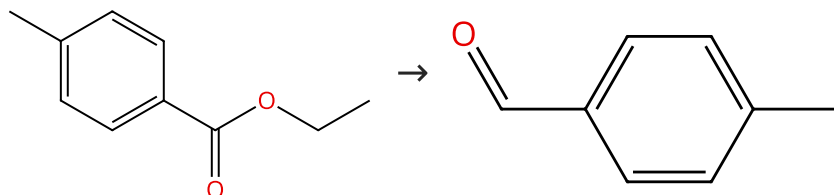
Suppliers (61)

Suppliers (83)

31-614-CAS-33829810	Steps: 1 Yield: 99%	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives
1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C		By: Young Kim, So; et al
1.2 10 min, 0 °C		ChemistrySelect (2022), 7(30), e202202351.
1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C		
Experimental Protocols		

Scheme 12 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (66)

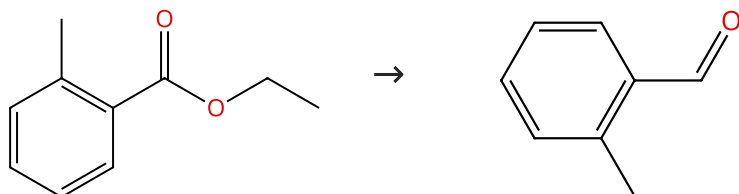
Suppliers (104)

31-614-CAS-33829848	Steps: 1 Yield: 99%	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives
1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C		By: Young Kim, So; et al
1.2 10 min, 0 °C		ChemistrySelect (2022), 7(30), e202202351.
1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C		
Experimental Protocols		

31-517-CAS-8440479	Steps: 1 Yield: 99%	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters
1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C		By: Jeon, Ah Ram; et al
1.2 0 °C; 10 min, 0 °C		Tetrahedron (2014), 70(29), 4420-4424.
1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C		
1.4 Reagents: Hydrochloric acid Solvents: Water		
Experimental Protocols		

Scheme 13 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (60)

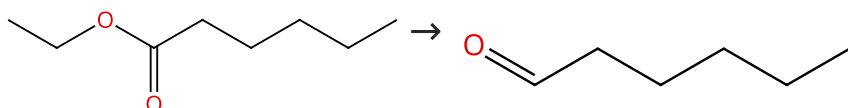
Suppliers (87)

31-614-CAS-33829842	Steps: 1 Yield: 99%	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives
1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C		By: Young Kim, So; et al
1.2 10 min, 0 °C		ChemistrySelect (2022), 7(30), e202202351.
1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C		
Experimental Protocols		

31-517-CAS-6306377 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.
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Scheme 14 (1 Reaction)

Steps: 1 Yield: 99%



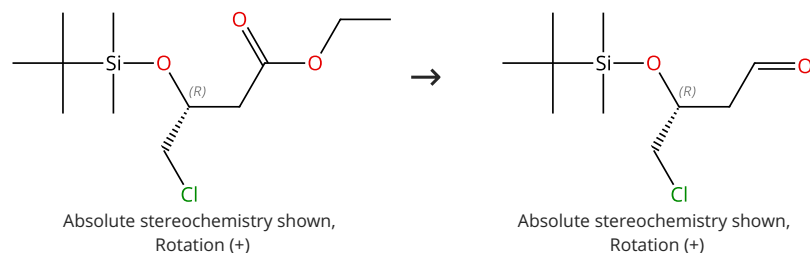
Suppliers (68)

Suppliers (68)

31-517-CAS-6583542 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Aluminate(1-), hydro(methanolato)bis(2-methyl propyl)-, lithium (1:1), (<i>T</i>-4)- Solvents: Tetrahydrofuran, Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.
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Scheme 15 (1 Reaction)

Steps: 1 Yield: 99%



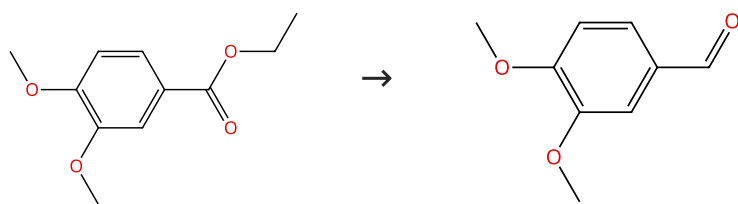
Suppliers (2)

Suppliers (11)

31-614-CAS-24777601 Steps: 1 Yield: 99% 1.1 Reagents: Diisobutylaluminum hydride Catalysts: Water Solvents: Toluene; 1 h, 78 °C 1.2 Reagents: Methanol; rt 1.3 Reagents: Potassium sodium tartrate Solvents: Water; 2 h, rt	Formal synthesis of Gram-negative antibiotic negamycin By: Zhang, Shiju; et al Youji Huaxue (2020), 40(2), 521-527.
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Scheme 16 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (62)

Suppliers (122)

31-517-CAS-2319243

Steps: 1 Yield: 99%

Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters

By: Jeon, Ah Ram; et al

Tetrahedron (2014), 70(29), 4420-4424.

1.1 **Reagents:** Morpholine, Diisobutylaluminum hydride
Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C

1.2 0 °C; 10 min, 0 °C

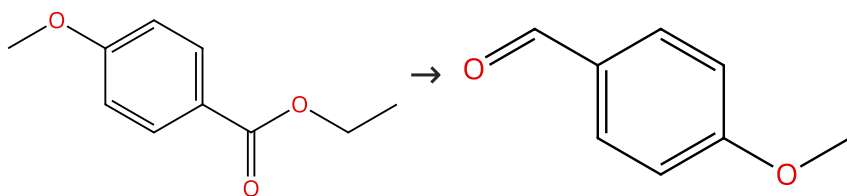
1.3 **Reagents:** Diisobutylaluminum hydride
Solvents: Hexane; 10 min, 0 °C

1.4 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (63)

Suppliers (128)

31-614-CAS-33829846

Steps: 1 Yield: 99%

Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives

By: Young Kim, So; et al

ChemistrySelect (2022), 7(30), e202202351.

1.1 **Reagents:** Morpholine, Diisobutylaluminum hydride
Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C

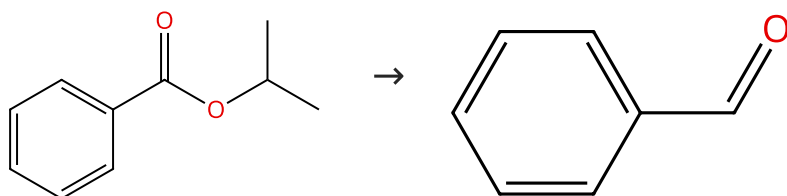
1.2 10 min, 0 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water; 0 °C

Experimental Protocols

Scheme 18 (2 Reactions)

Steps: 1 Yield: 99%



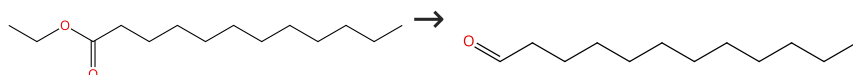
Suppliers (67)

Suppliers (69)

31-614-CAS-33829806 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C 1.2 10 min, 0 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C Experimental Protocols	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives By: Young Kim, So; et al ChemistrySelect (2022), 7(30), e202202351.
31-517-CAS-6912712 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



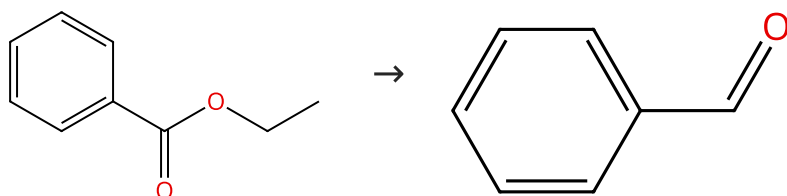
Suppliers (80)

Suppliers (76)

31-517-CAS-8719062 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Aluminate(1-), hydro(methanolato)bis(2-methyl propyl)-, lithium (1:1), (7-4)- Solvents: Tetrahydrofuran, Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.
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Scheme 20 (2 Reactions)

Steps: 1 Yield: 99%



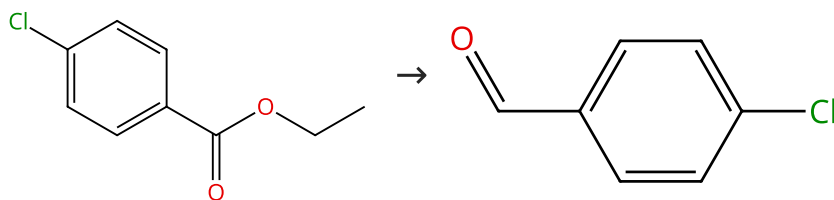
Suppliers (90)

Suppliers (69)

31-614-CAS-33829807 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C 1.2 10 min, 0 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C Experimental Protocols	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives By: Young Kim, So; et al ChemistrySelect (2022), 7(30), e202202351.
31-517-CAS-4774785 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.

Scheme 21 (2 Reactions)

Steps: 1 Yield: 99%



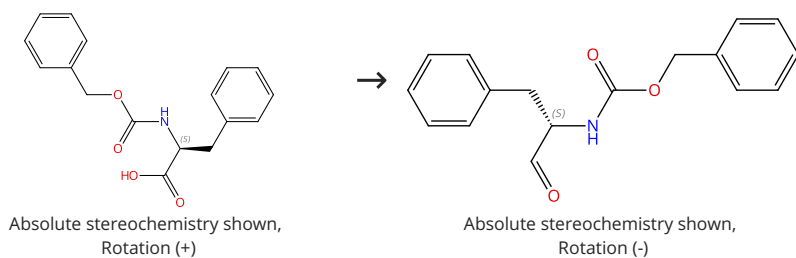
Suppliers (71)

Suppliers (99)

31-614-CAS-33829823 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C 1.2 10 min, 0 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C Experimental Protocols	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives By: Young Kim, So; et al ChemistrySelect (2022), 7(30), e202202351.
31-517-CAS-13310965 Steps: 1 Yield: 99% 1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 0 °C; 3 h, 0 °C 1.2 0 °C; 10 min, 0 °C 1.3 Reagents: Diisobutylaluminum hydride Solvents: Hexane; 10 min, 0 °C 1.4 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Mild and direct conversion of esters to morpholine amides using diisobutyl(morpholino)aluminum: application to efficient one-pot synthesis of ketones and aldehydes from esters By: Jeon, Ah Ram; et al Tetrahedron (2014), 70(29), 4420-4424.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (97)

Suppliers (45)

31-517-CAS-3014271

Steps: 1 Yield: 99%

A rapid and efficient one-pot method for the reduction of N-protected α -amino acids to chiral α -amino aldehydes using C DI/DIBAL-H

By: Ivkovic, Jakov; et al

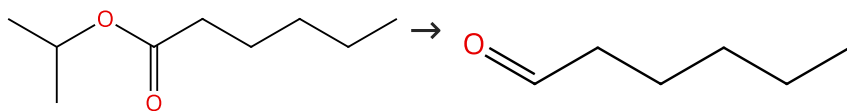
Organic & Biomolecular Chemistry (2015), 13(42), 10456-10460.

- 1.1 **Reagents:** 1,1'-Carbonyldiimidazole
Solvents: Dichloromethane; 60 min, 0 °C; 15 min, 0 °C → - 78 °C
- 1.2 **Reagents:** Diisobutylaluminum hydride
Solvents: Toluene; 110 min, 0 °C; 30 - 60 min, - 78 °C
- 1.3 **Reagents:** Ethyl acetate; - 78 °C

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (48)

Suppliers (68)

31-517-CAS-10431636

Steps: 1 Yield: 98%

Lithium diisobutyl-*t*-butoxyaluminum hydride, a new and efficient reducing agent for the conversion of esters to aldehydes

By: Kim, Min Sung; et al

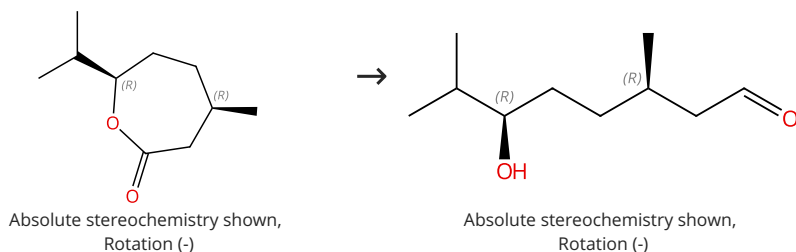
Tetrahedron Letters (2007), 48(29), 5061-5064.

- 1.1 **Reagents:** Aluminate(1-), hydro(2-methyl-2-propanolato)bis(2-methylpropyl)-, lithium (1:1), (*T*-4)-
Solvents: Tetrahydrofuran, Hexane; 1 h, 0 °C
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 98%

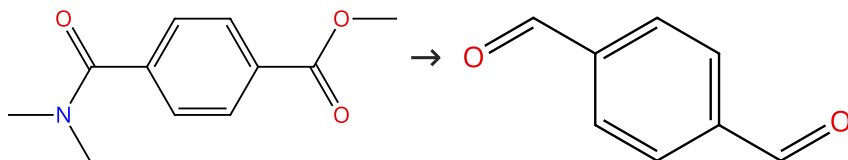


Suppliers (2)

31-517-CAS-5572108	Steps: 1 Yield: 98%	Low-temperature reduction of acyclic (-)-mentholactone derivatives with diisobutylaluminum hydride in methylene chloride
1.1 Reagents: Diisobutylaluminum hydride Solvents: Dichloromethane, Toluene; rt → -70 °C; 3 h, -70 °C		By: Ishmuratov, G. Yu.; et al
1.2 Reagents: Water Solvents: Tetrahydrofuran		Russian Journal of Organic Chemistry (2015), 51(7), 947-950.
Experimental Protocols		

Scheme 25 (2 Reactions)

Steps: 1 Yield: 98%



Suppliers (46)

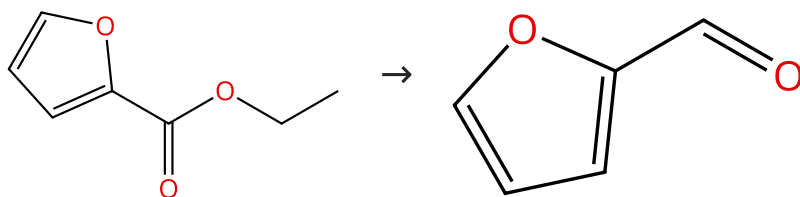
Suppliers (83)

31-614-CAS-33829803	Steps: 1 Yield: 98%	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives
No Data Available		By: Young Kim, So; et al
Experimental Protocols		ChemistrySelect (2022), 7(30), e202202351.

31-614-CAS-33829863	Steps: 1 Yield: 98%	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives
1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C		By: Young Kim, So; et al
1.2 10 min, 0 °C		ChemistrySelect (2022), 7(30), e202202351.
1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C		
Experimental Protocols		

Scheme 26 (1 Reaction)

Steps: 1 Yield: 98%



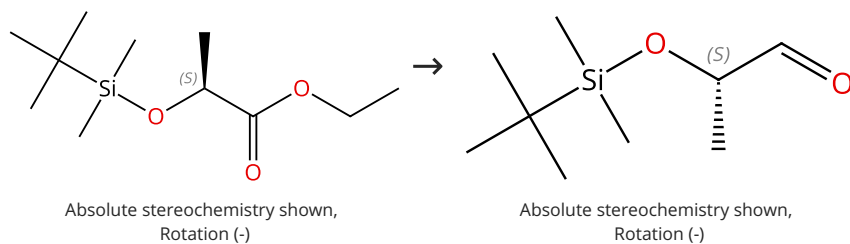
Suppliers (71)

Suppliers (81)

31-614-CAS-33829854	Steps: 1 Yield: 98%	Diisobutyl(morpholino)aluminum-Enabled Partial Reductions with DIBALH: Highly Practical Synthesis of Aldehydes from Carboxylic Acid Derivatives
1.1 Reagents: Morpholine, Diisobutylaluminum hydride Solvents: Tetrahydrofuran, Hexane; 2 min, 0 °C; 30 min, 0 °C		By: Young Kim, So; et al
1.2 10 min, 0 °C		ChemistrySelect (2022), 7(30), e202202351.
1.3 Reagents: Hydrochloric acid Solvents: Water; 0 °C		
Experimental Protocols		

Scheme 27 (1 Reaction)

Steps: 1 Yield: 98%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (-)

Suppliers (51)

Suppliers (47)

31-517-CAS-6263544

Steps: 1 Yield: 98%

Diisobutylaluminum Hydride Reductions Revitalized: A Fast, Robust, and Selective Continuous Flow System for Aldehyde Synthesis

By: Webb, Damien; et al

Organic Letters (2012), 14(2), 568-571.

1.1 Reagents: Diisobutylaluminum hydride

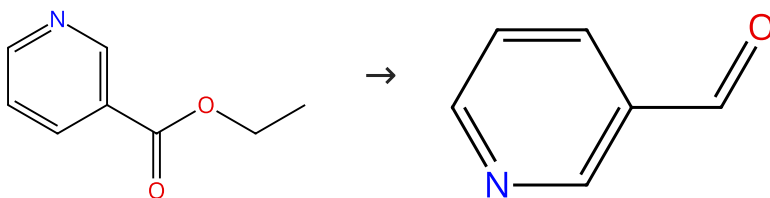
Solvents: Toluene; 13.9 s, -42 °C

1.2 Reagents: Methanol

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (95)

Suppliers (92)

31-517-CAS-5659356

Steps: 1 Yield: 97%

Application of Flow Chemistry to the Selective Reduction of Esters to Aldehydes

By: Munoz, Juan de M.; et al

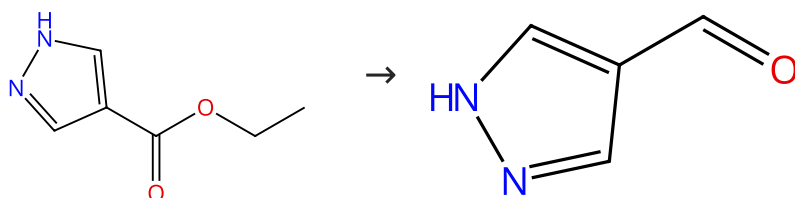
European Journal of Organic Chemistry (2012), 2012(2), 260-263, S260/1-S260/8.

1.1 Reagents: Aluminate(1-), hydro(2-methyl-2-propanolato)bis(2-methylpropyl)-, lithium (1:1), (7-4)-
Solvents: Toluene; 0 °C

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (89)

Suppliers (75)

31-517-CAS-1633879

Steps: 1 Yield: 97%

Application of Flow Chemistry to the Selective Reduction of Esters to Aldehydes

By: Munoz, Juan de M.; et al

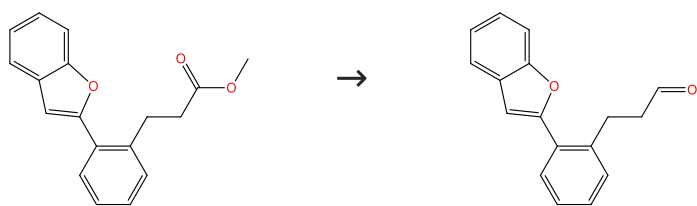
European Journal of Organic Chemistry (2012), 2012(2), 260-263, S260/1-S260/8.

1.1 Reagents: Aluminate(1-), hydro(2-methyl-2-propanolato)bis(2-methylpropyl)-, lithium (1:1), (7-4)-
Solvents: Toluene; 0 °C

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 97%



31-517-CAS-7623636

Steps: 1 Yield: 97%

Mechanistic Insights into the Post-Cyclization Isomerization in Gold-Catalyzed 7-exo-dig-Hydroarylations

By: Pflaesterer, Daniel; et al

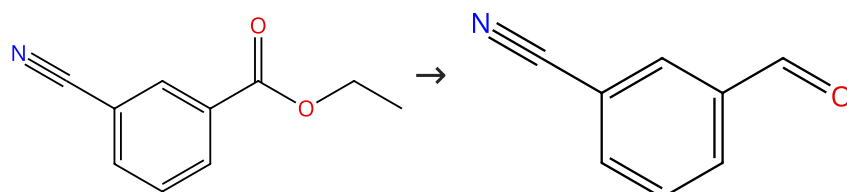
Chemistry - A European Journal (2015), 21(32), 11585-11589.

1.1 **Reagents:** Diisobutylaluminum hydride
Solvents: Dichloromethane, Hexane; 2 h, -90 °C

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (56)

Suppliers (87)

31-517-CAS-10468856

Steps: 1 Yield: 97%

Application of Flow Chemistry to the Selective Reduction of Esters to Aldehydes

By: Munoz, Juan de M.; et al

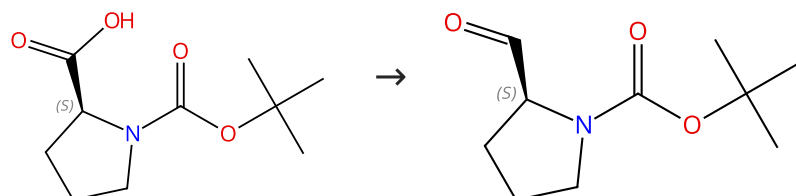
European Journal of Organic Chemistry (2012), 2012(2), 260-263, S260/1-S260/8.

1.1 **Reagents:** Aluminate(1-), hydro(2-methyl-2-propanolato)bis(2-methylpropyl)-, lithium (1:1), (7-4)-
Solvents: Toluene; 0 °C

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 97%



Absolute stereochemistry shown,
 Rotation (-)

Suppliers (106)

Absolute stereochemistry shown,
 Rotation (-)

Suppliers (79)

31-517-CAS-13174960

Steps: 1 Yield: 97%

A rapid and efficient one-pot method for the reduction of N-protected α-amino acids to chiral α-amino aldehydes using C DI/DIBAL-H

By: Ivkovic, Jakov; et al

Organic & Biomolecular Chemistry (2015), 13(42), 10456-10460.

1.1 **Reagents:** 1,1'-Carbonyldiimidazole
Solvents: Dichloromethane; 60 min, 0 °C; 15 min, 0 °C → -78 °C

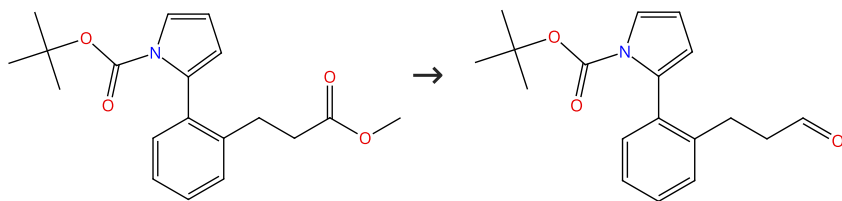
1.2 **Reagents:** Diisobutylaluminum hydride
Solvents: Toluene; 110 min, -78 °C; 30 - 60 min, -78 °C

1.3 **Reagents:** Ethyl acetate; -78 °C

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 96%



31-517-CAS-891428

Steps: 1 Yield: 96%

Mechanistic Insights into the Post-Cyclization Isomerization in Gold-Catalyzed 7-exo-dig-Hydroarylations

1.1 Reagents: Diisobutylaluminum hydride

Solvents: Dichloromethane, Toluene; 2 h, -90 °C

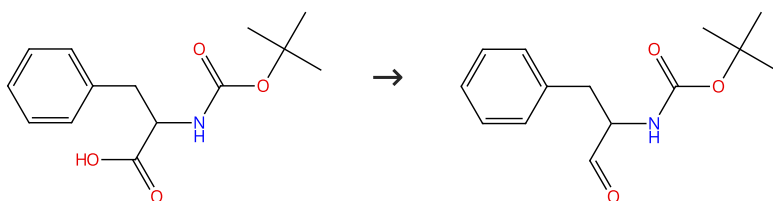
By: Pflaesterer, Daniel; et al

Experimental Protocols

Chemistry - A European Journal (2015), 21(32), 11585-11589.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (67)

Suppliers (45)

31-517-CAS-2236922

Steps: 1 Yield: 96%

A rapid and efficient one-pot method for the reduction of N-protected α -amino acids to chiral α -amino aldehydes using C DI/DIBAL-H

1.1 Reagents: 1,1'-Carbonyldiimidazole

Solvents: Dichloromethane; 60 min, 0 °C; 15 min, 0 °C $\rightarrow$ -78 °C

1.2 Reagents: Diisobutylaluminum hydride

Solvents: Toluene; 110 min, -78 °C; 30 - 60 min, -78 °C

1.3 Reagents: Ethyl acetate; -78 °C

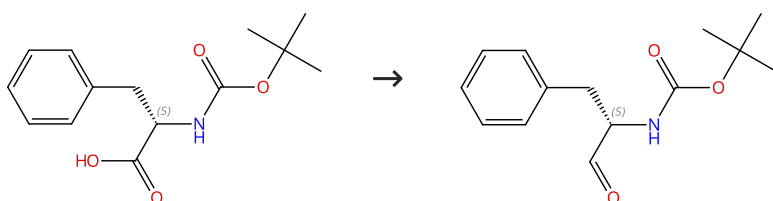
By: Ivkovic, Jakov; et al

Organic & Biomolecular Chemistry (2015), 13(42), 10456-10460.

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 96%

Absolute stereochemistry shown,
Rotation (+)

Suppliers (101)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (63)

31-517-CAS-414508

Steps: 1 Yield: 96%

A rapid and efficient one-pot method for the reduction of N-protected α -amino acids to chiral α -amino aldehydes using C DI/DIBAL-H

1.1 Reagents: 1,1'-Carbonyldiimidazole

Solvents: Dichloromethane; 60 min, 0 °C; 15 min, 0 °C $\rightarrow$ -78 °C

1.2 Reagents: Diisobutylaluminum hydride

Solvents: Toluene; 110 min, -78 °C; 30 - 60 min, -78 °C

1.3 Reagents: Ethyl acetate; -78 °C

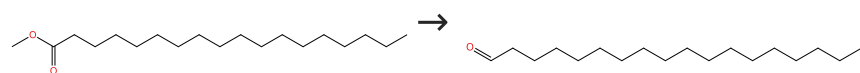
By: Ivkovic, Jakov; et al

Organic & Biomolecular Chemistry (2015), 13(42), 10456-10460.

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (107)

Suppliers (60)

31-517-CAS-22769972

Steps: 1 Yield: 95%

Influence of Some Factors on the Progress of a New Reaction in the Chemistry of Organoaluminum Compounds

1.1 **Reagents:** Diisobutylaluminum hydride
Solvents: Dichloromethane, Toluene, Tetrahydrofuran, Water;
-70 °C; 3 h, -70 °C

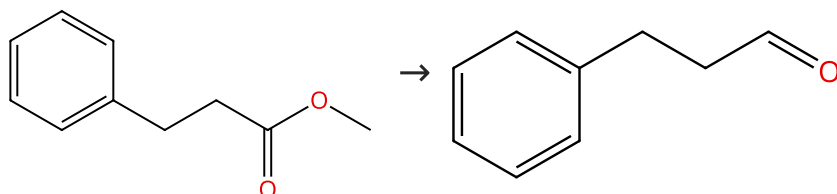
By: Yakovleva, M. P.; et al

Russian Journal of Organic Chemistry (2020), 56(8), 1353-1358.

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (74)

Suppliers (90)

31-517-CAS-8122574

Steps: 1 Yield: 95%

Diisobutylaluminum Hydride Reductions Revitalized: A Fast, Robust, and Selective Continuous Flow System for Aldehyde Synthesis

1.1 **Reagents:** Diisobutylaluminum hydride
Solvents: Toluene; 6.6 s, -42 °C

By: Webb, Damien; et al

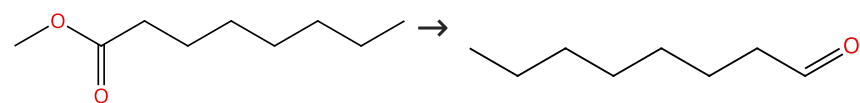
Organic Letters (2012), 14(2), 568-571.

1.2 **Reagents:** Methanol

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (102)

Suppliers (81)

31-517-CAS-10246782

Steps: 1 Yield: 95%

Diisobutylaluminum Hydride Reductions Revitalized: A Fast, Robust, and Selective Continuous Flow System for Aldehyde Synthesis

1.1 **Reagents:** Diisobutylaluminum hydride
Solvents: Toluene; 15.2 s, -78 °C

By: Webb, Damien; et al

Organic Letters (2012), 14(2), 568-571.

1.2 **Reagents:** Methanol

Experimental Protocols